MILITARY SPECIFICATION SHEET

ELECTRON TUBE, CATHODE RAY

TYPES 2BP1 AND 2BP11

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Electrostatic deflection and focus

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	Type phosphor	Ef	Ec1	ed	Eb1	Eb2	Light output	Rg	Zd	Ehk	Alt
Unit:		V	Vdc	v	Vdc	Vdc	ftL	Meg	Meg	Vdc	ft
Maximum:	P1, P11	6.9	-200	550	1,100	2,750	---	1.5	1.0	±125	30,000
Minimum:	P1	5.7	0	---	---	500(Note 2)	4.0	---	---	---	---
Minimum:	P11	5.7	0	---	---	1,100	2.25	---	---	---	---
TEST CONDITIONS:	P1, P11	6.3	Adj	---	Focus	1,000	---	---	---	---	---

GENERAL:

Qualification - Required (see note 1)

Ⓓ. denotes changes

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METHOD	REQUIREMENT OR TEST	TYPES	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
5111	Vibration	P1, P11	-		Width	---	1.0	mm
1002	Barometric pressure, reduced	P1, P11	6		---	---	---	---
5101	Neck and bulb alinement (electrostatic types)	P1, P11	-		Diam.	---	1.63	inch
(D)5216	Cathode illumination	P1, P11	-		---	---	---	---
(D)5246	Focusing voltage at modulation condition	P1, P11	-		Ebf	150	280	Vdc
(D)5248	Deflection-factor uniformity	P1, P11	-		---	---	---	---
1331	Direct-interelectrode capacitance	P1, P11	-	g1 to all D1 to D2 D3 to D4 D1 to all D2 to all D3 to all D4 to all	Cg1 C1D2 C3D4 CD1 CD2 CD3 CD4	--- --- --- --- --- --- ---	11.0 4.0 4.0 15.0 11.0 10.0 11.0	pF pF pF pF pF pF pF
1141	Pressure (implosion)	P1, P11	-		---	---	---	---
(D)1216	Base material insulating quality	P1, P11	-		---	---	---	---
	Quality conformance inspection, part 1							
5201	Voltage breakdown	P1, P11	-		---	---	---	---
5201	Voltage breakdown (electrostatic types)	P1, P11	-		---	---	---	---
5206	Gas "cross"	P1 P11	7 7	Light = 4.0 ftL Light = 2.25 ftL	--- ---	--- ---	--- ---	--- ---
5106	Bulb, screen, and faceplate quality	P1, P11			---	---	---	---
5221	Light output	P1 P11	7 5,7		Light Light	4.0 2.25	--- ---	ftL ftL
5223	Modulation	P1 P11	- -	Light = 4.0 ftL Light = 2.25 ftL	ΔEc1 ΔEc1	--- ---	48 48	Vdc Vdc
5231	Spot position (electrostatic deflection)	P1, P11	-		---	---	10	mm
5231	Spot displacement (leakage)	P1, P11	-		Displ	---	6	mm
5241	Grid cutoff voltage	P1, P11	-		Ec1	---	-67.5	Vdc
	Quality conformance inspection, part 2							
1301	Heater current	P1, P11	-		If	540	660	mA
5201	Electrode current (anode No. 1)	P1, P11	-	Ec1 = 0	Ib1	-15	10	uAdc
5201	Electrode current (cathode)	P1 P11	- -	Light = 4.0 ftL Light = 2.25 ftL	Ik Ik	--- ---	1,000 1,000	uAdc uAdc
5101	Base alinement (electrostatic types)	P1, P11	-	3D4; pin No. 1	---	---	---	---

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METHOD	REQUIREMENT OR TEST	TYPES	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued							
5101	Angle between traces	P1, P11	-		Angle	87	93	Degrees
5101	Neck and base alignment (electrostatic types)	P1, P11	-		---	---	---	---
5216	Stray light emission	P1, P11	-		---	---	---	---
5226	Line width "A" (electrostatic types)	P1 P11	3 -	Light = 4.0 ftL Light = 2.25 ftL	Width Width	--- ---	0.45 0.45	cm cm
5226	Line width "B" (electrostatic deflection)	P1 P11	4 4	Light = 4.0 ftL Light = 2.25 ftL	Width Width	--- ---	0.45 0.45	cm cm
---	Focusing voltage	P1 P11	- -	Light = 4.0 ftL Light = 2.25 ftL	Eb1 Eb1	150 150	280 280	Vdc Vdc
5248	Deflection factor (1D2)	P1, P11	-		DF	115	155	Vdc/in.
(D)1111	Base pin solder depth	P1, P11	-		---	---	---	---
5248	Deflection factor (3D4)	P1, P11	-		DF	74	100	Vdc/in.
5251	Heater-cathode leakage current	P1, P11	-		---	---	---	---
5251	Grid No. 1 leakage current	P1, P11	-		---	---	---	---
5251	Anode No. 2 leakage current	P1, P11	-		---	---	---	---
1101	Secureness of base, cap, or insert	P1, P11	-		---	---	---	---
(D)1105	Permanence of marking	P1, P11	-		---	---	---	---
	Quality conformance inspection, part 3							

---	Life test	P1	-	Group C;
				t = 500 hours (min);
				light = 4.0 ftL;
				Eb2 = 2,750 Vdc
					---	---	---	---

		P11	-	Group C;
				t = 500 hours (min);
				light = 2.25 ftL;
				Eb2 = 2,750 Vdc
					---	---	---	---

---	Life-test end points:							
5226	Line width "A"	P1	-	Light = 3.4 ftL	Width	---	0.45	cm
5226	Line width "B"	P1	-	Light = 3.4 ftL	Width	---	0.45	cm
5223	Modulation	P1	-	Light = 3.4 ftL	ΔEC	---	48	Vdc
5226	Line width "A"	P11	-	Light = 2.25 ftL	Width	---	0.45	cm
5226	Line width "B"	P11	-	Light = 2.25 ftL	Width	---	0.45	cm
5223	Modulation	P11	-	Light = 2.25 ftL	ΔEc	---	48	Vdc

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NOTES:

1. The construction of this gun shall be of the zero Ibl type. The following information and materials are to be forwarded with the four regular samples when application for qualification approval is made:

   1 each gun drawings with significant dimensions.
   1 each sample of gun to be used in manufacture of tubes.

2. This value is recommended only for low-velocity deflecting and low-ambient light levels.

3. The deflecting plates shall be returned to anode No. 2 through a minimum of 2.5 megohm resistors. Ib3, the beam current, shall be set at 50 uAdc. The high-frequency scanning shall be applied to the deflecting plates nearest the screen and the amplitude shall be adjusted to give a line length of approximately 90 percent of the maximum tube diameter. The low-frequency scanning amplitude shall be expanded to approximately 90 percent of the maximum tube diameter in the direction perpendicular to the direction of high-frequency scanning. Readjustment may be made for best overall focus. The tubes shall be observed for deflection defocusing, astigmatism, or spot ellipticity observable to the eye as evidenced by fuzziness due to lack of sharpness of trace (usually around edges), bow-tieing (irregular widths of any single line when observed at different points), bowing of trace other than that normally caused by curvature of bulb. This test for focus is to be made in addition to the line width measurements.

4. The same conditions shall be set up as described in note 3 except that the connection of deflecting elements to the low- and high-frequency scanning supplies shall be interchanged and the amplitudes adjusted to 90 percent of the maximum tube diameter in both directions without any adjustment of focus from note 3 conditions. An examination for defocusing, astigmatism, or spot ellipticity shall be made as in note 3.

5. As measured by a 2x2-inch raster using a type 3 photronic cell without eye correction, calibrated in footcandles of illumination from a light source having color temperature of 2,700°K.

6. The test is made with maximum voltage (Eg1 maximum negative voltage) applied to the base pins, and deflection electrodes if applicable, only; and pressure of 30,000 feet (225 mmHg). Connections should be made in a manner that does not degrade the tube's electrical voltage breakdown characteristics. Satisfactory operation is the absence of arc-over and corona.

7. This test to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MU, MI
Air Force - 17, 80
DSA - ES

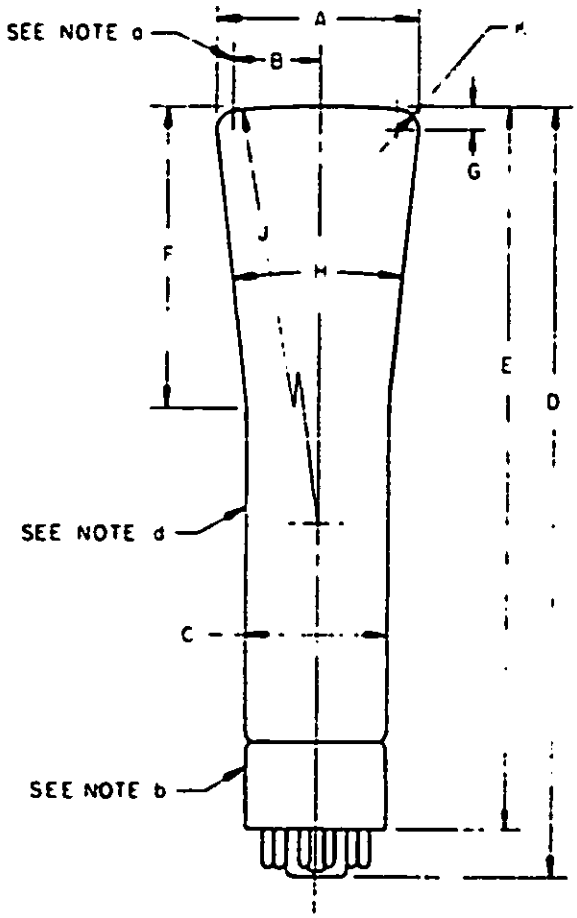
User activities:
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

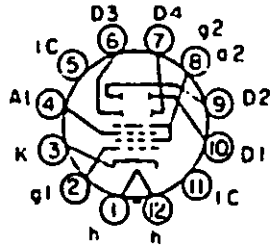
Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	1.938 (49.23)	2.063 (52.40)
B	.875 (22.23)	
C	1.312 (33.43)	1.438 (36.53)
D	7.438 (188.93)	7.812 (198.42)
E	6.938 (176.23)	7.312 (185.72)
Reference dimensions (see note c)		
F	3.062 (77.77)	
G	.225 (5.72)	
H	12°37'	
J	8.00 R	
K	.188 (4.78) R	



PIN CONNECTIONS

NOTES:
a. The minimum useful screen radius.
b. The base shall be a small shell duodecal 12 pin or small shell duodecal minus pins 5 and 11 (JEDEC B10-75).
c. Reference dimensions are for information only and are not required for inspection purposes.
d. The bulb shall be a J16A type. Either clear or filter glass may be used.

FIGURE 1. Outline drawing of electron tube types 2BP1, 2BP11.

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